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# PHYSICS

## JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

# Work, Power & Energy

(English Medium)

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# WORK, POWER & ENERGY

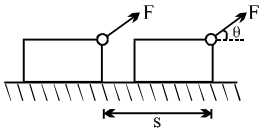
# KEY CONCEPT

**Work (W) :**

The **work W** done by a constant force F when its point of application undergoes a displacement s is defined as

$$W = F.s = Fs\cos\theta$$

where θ is the angle between F and s. Work is a scalar quantity and its SI units is N-m or joule (J).




**Note:** Only the component (F cos θ) of the force F which is along the displacement contributes to the work done.

If $F = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$ and $s = \Delta x\hat{i} + \Delta y\hat{j} + \Delta z\hat{k}$

then $W = \vec{F}\cdot\vec{s} = F_x\Delta x + F_y\Delta y + F_z\Delta z$

- **Work done by a Variable Force :** When the magnitude and direction of a force varies with position, The work done by such a force for an infinitesimal displacement ds is given by

$$dW = \vec{F}\cdot d\vec{s}$$

In terms of rectangular components,

$$W_{AB} = \int_{X_A}^{X_B} F_x dx + \int_{Y_A}^{Y_B} F_y dy + \int_{Z_A}^{Z_B} F_z dz$$

- **Work Done by a Spring Force :** The work done by the spring force for a displacement from $x_i$ to $x_f$ is given by

$$W_s = -\frac{1}{2}k\left(x_f^2 - x_i^2\right)$$

- **Work Energy theorem :**

Work done on a body can produce a change in its kinetic energy. Work is required to produce motion and it is also required to destroy motion.

$$W = \Delta K = K_f - K_i$$

- **Conservative Force :** The force which does work in complete independence of the path followed the body is called a conservative force. The gravitational force, spring force and electrostatic force are the examples of conservative forces.

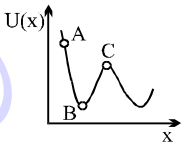
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- **Non-Conservative Force :** The work done by a non-conservative force not only depends on the initial and final positions but also on the path followed. The common examples of such forces are : frictional force and drag force of fluids.

- **Potential Energy :** The potential energy is defined only for conservative forces.

$$U_B - U_A = -\int_A^B F_c.ds$$

- Conservative force : $F_c = -\frac{dU}{dx}$




At equilibrium, $\frac{dU}{dx} = 0$

The point B is the position of stable equilibrium, because $\frac{d^2U}{dx^2} > 0$

The point C is the position of unstable equilibrium, because $\frac{d^2U}{dx^2} < 0$

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